

Multiple Choice (10 marks)

1. Which of the following conditions does not show multifactorial inheritance?
 - (a) Pyloric stenosis
 - (b) Schizophrenia
 - (c) Spina bifida (neural tube defects)
 - (d) Marfan syndrome
2. Familial hypercholesterolaemia is caused by mutations in the gene which encodes what?
 - (a) High density lipoprotein
 - (b) HMG-CoA reductase
 - (c) Low density lipoprotein
 - (d) Low density lipoprotein receptor
3. Which of the following is the most important for controlling disease caused by norovirus?
 - (a) A norovirus vaccine
 - (b) Attention to handwashing and hygiene
 - (c) Quarantine
 - (d) New antiviral drugs
4. Which of the following karyotypes is diagnostic of Down syndrome
 - (a) 46,XX,der(14;21)(q 10;q 10)pat+21
 - (b) 47,XY,+13
 - (c) 45,XX,rob,(14;21)(q 10;q 10)
 - (d) 46,XY,t(2;3)(q 21;q 12)
5. Which virus infection was not first identified by clinical observation?
 - (a) HIV
 - (b) Rubella teratogenicity
 - (c) Microcephaly and Zika
 - (d) Hepatitis C

True / False (10 marks)

Answer True or False

6. True or False: The aorta is responsible for transporting air to the lungs from the atmosphere.
7. True or False: Speaking loudly into their best ear is the most effective way to ensure older adults with hearing loss understand you.
8. True or False: A homeotic mutation substitutes one body part for another during an organism's development.
9. True or False: Adenoviruses used in gene therapy have a low risk of insertional mutagenesis, making them safer for therapeutic applications.
10. True or False: Recombinant alpha-iduronidase is primarily used to treat Fabry disease, which is caused by a different enzyme deficiency.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors that may contribute to higher rates of depression in young survivors following a loss, and analyze the implications for their mental health and support needs.
12. Discuss the significance of clonal expansion in the adaptive immune system and analyze how it contributes to the effectiveness of immune responses against pathogens.

End of Paper